

Corporate DIY: Jet Fuel for Innovation

Industry leaders are encouraging a DIY culture amongst their employees. We took a peak at some of the DIY stuff displayed at ARM's recent tech fest and discussed how DIY helps a company grow...

ARM was all ablaze with excitement as it celebrated ingenuity and innovation at the recently held TechnoFest 2013. Krishna Kumar Ranganathan, principal engineer, ARM India, had some interesting stuff on platform stabilisation and demonstrated a miniature 'Segway' kind of mobile device at the event. "I have a bigger prototype at home and I am keen on building a cost-effective human transportation device," he said.

Another group of employees showcased a Smart Aquarium project. Their product involved an Android app that could help one remotely monitor an aquarium and program the feeding time.

"ARM has always encouraged employees to innovate and explore the world of technology beyond their regular daily routines. The company provides mbeds to employees to conduct their projects," says Ranganathan.

The mbed kit is a single-board microcontroller with associated tools for programming the device. It is a rapid prototyping platform that helps complete microprocessor-based system prototypes in a significantly short period of time. The mbed project was initiated by two ARM employees, and later became an official research project within ARM. It is now run and maintained by ARM to help MCU Partners provide their customers with the best way to prototype designs using their microcontrollers. mbed began with NXP LPC 1768 (ARM CortexM3) but now includes NXP LPC1114 (ARM CortexM0) and FSCAL Freedom Board (ARM CortexM0+). ARM also collaborates with universities and high-schools and provides mbed

kits to students for learning and experiments.

Apart from mbed, if the employees need any other components or software for their projects, ARM readily funds the projects if it is satisfied with the purpose. For instance, DIY enthusiasts at ARM are now working on building a robot that can navigate its way through their office collecting coffee cups.

Ranganathan explains, "Let's say I'm at my desk, and I need a cup of coffee. I just have to go to the intranet and request a cup of coffee. The robot will recognise my login and hence know where I am. By default, it comes to my desk but I can also mention if I am elsewhere in the office. The robot goes to the vending machine and brings the cup of coffee that has already been prepared. The robot is basically a smart mobile platform with wheels. The idea is that the robot will navigate from place A to place B within the office avoiding all obstacles and knowing the destination as well. It can also go about collecting empty mugs."

DIY enthusiasts team at ARM is also interested in adding video conferencing capabilities to the robot.

Guru Ganesan, managing director of ARM India, explains, "DIY projects benefit the organisation as well as the individuals. These help individuals by encouraging them to think out of the box. Some of these projects might actually enhance daily operations, which is beneficial for the company. ARM has always encouraged such employee-led innovation ideas that help them co-create for a shared purpose."

we perceive DIY for engineering problems. It may not be always possible to build a big project from scratch to end in DIY mode. Deterred by this thought, people drop highly-technical projects and go for simple ones. However, the solution is to learn how to modularise or segment a big project to be able to take up a part of it in DIY mode," says Jayakrishnan.

He also feels that people should get out of the mindset that DIY must involve hardware development—developing an algorithm or finding solutions to mathematical bottlenecks can also be DIY activities!

"I think the largest pitfall is that you often think that if you have built something and have it sitting in front of you, people online would like it and you immediately have a product for sale. Building the prototype of a singular unique item is not necessarily the same as designing a product for mass market, nor does having an audience mean that people will pay the actual cost of production once the product hits the market," explains Sarafan.

Kuber adds, "India has a very

brilliant DIY community. But Indians have to nurture their documenting and publicity talents. A small desktop experiment, if video-documented well and uploaded to the correct social media with correct tutorial instructions, would be a great gift to the spirit of open source."

DIY in India

"India has a long tradition of ingenuity and a strong technical base. The DIY/maker ethic is tremendously strong here. We consistently see new and exciting projects from India," says Denmead.

Sarafan says, "Having never been to India, my observation of the DIY community in India can only be based on what I observe on Instructables. However, that said, it seems to be pretty vibrant. We run hundreds of contests on our site every year for DIY projects, and India ranks within the top 10 countries throughout the world in terms of the number of winners. I'm also impressed by the number of children from India who regularly contact me for help in their electronics projects.

There seems to be a thriving DIY scene, especially amongst younger people."

That said, the people actually getting their hands dirty have mixed views on the Indian DIY ecosystem. "DIY has never been big in India," moans Kashinath. "Not all schools can boast of a 'workshop' where students can get their hands dirty working with metal, wood, electronics, etc. There is no famous hobby shop that promotes DIY either. As a result, it just isn't something that kids are exposed to. Some TV channels and schools today are leading the effort in this, which is heartening to see. I was lucky to have a workshop in my school, and I do believe that it contributed significantly to my present-day interest in this area. That said, it is promising to see how things are evolving here. Information and ideas are accessible globally, hardware and prototyping costs are reducing, local communities are forming and the momentum is increasing." ●

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